

Fan Control Board Manual

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FIT Collaboration ALICE CERN

General Description

Fan Control Board FCB is a custom electronic board for using in FIT detector of ALICE experiment at CERN. It is designed for controlling custom air cooling panel installed in Wiener VME crate. This board provides information about fan rotation speeds and magnetic environment status inside rack. It also allows switching power for external devices.

Keywords: INR RAS, FIT Collaboration, ALICE, CERN

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Board Description

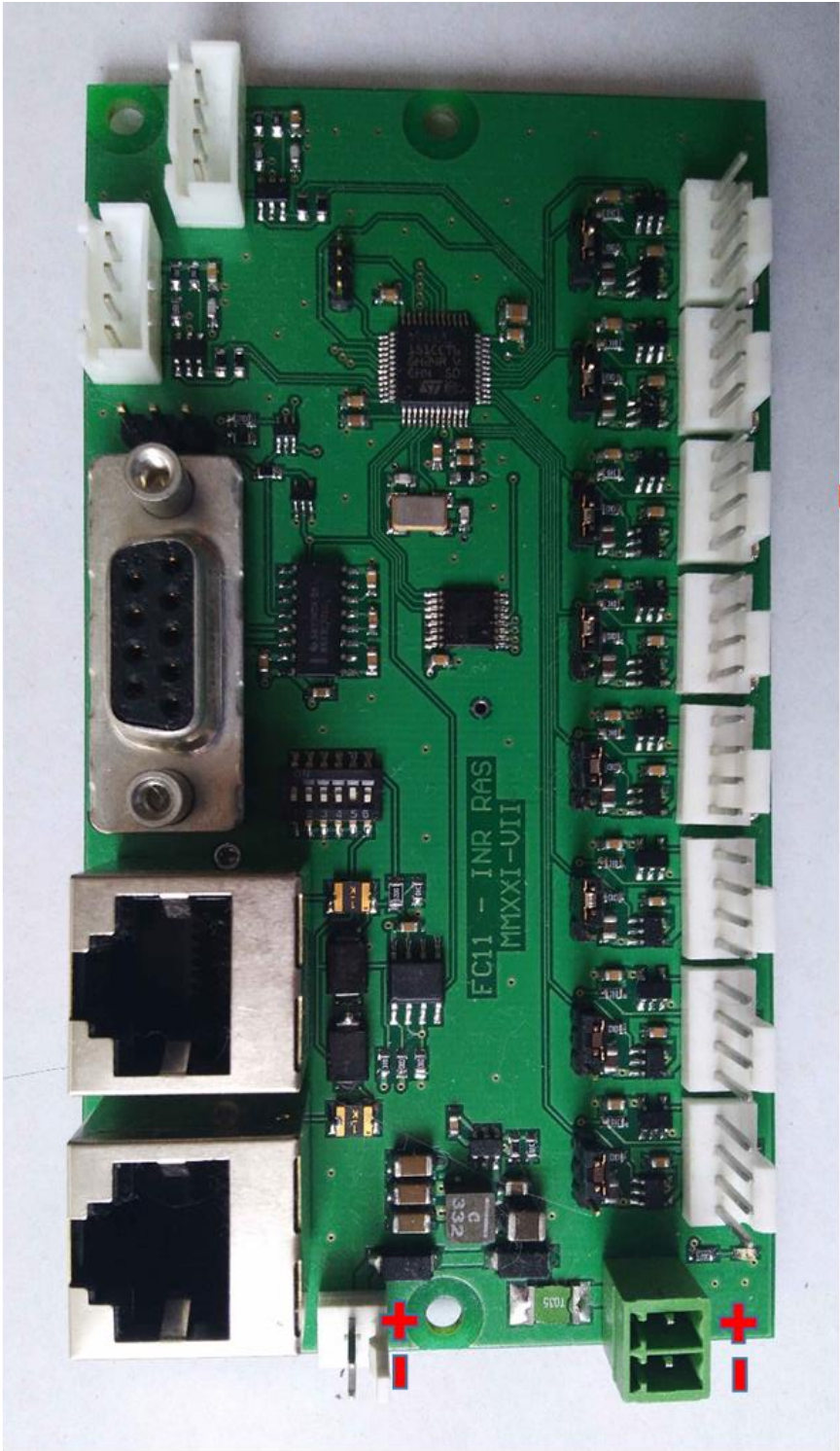
Main features

- RS-232 communication.
- RS-485 communication.
- Configuration switch: device address – from 0 to 31, switchable RS-485 line termination.
- 5 Fan outputs.
- 2 relay outputs.
- 2 3-axis magnetic sensors up to 200 mT linear range.
- 12V main power. Maximum overall current 2A.
- 12V auxiliary power. Current less than 100 mA.

Magnetic
Sensor
Connectors

RS-232

RS-485 in & out



Reserved
Relay L
Relay M

Fan1 Fan2 Fan3 Fan4 Fan5

12V AUX

12V from
Crate

Configuration switch

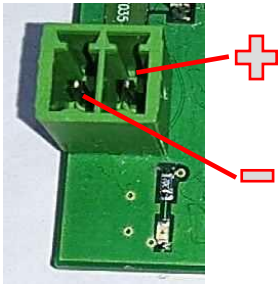


1	2	3	4	5	6
A4	A3	A2	A1	A0	Termination

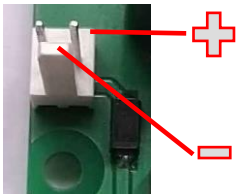
Connectors

Power.

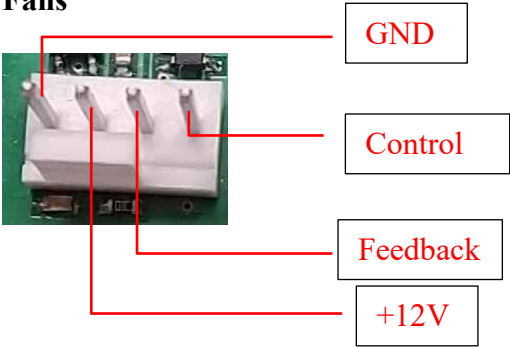
12V main power connector. Supply current through this connector is about 1.5A



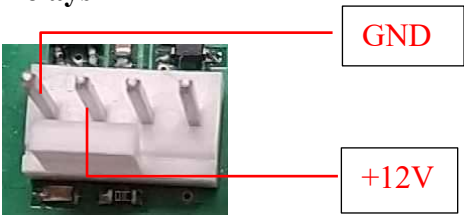
12V auxiliary power connector. When the board is power only via this connector, FCB can only respond to communication commends and monitor magnetic sensors. Fans and controlled switches are in unpowered state. Supply current is several tens of mA.



Fans



Relays



Communication

Two RS-485 connectors.



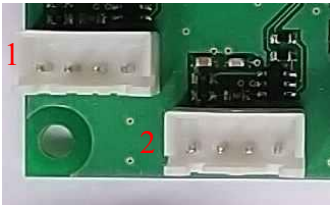
8	7	6	5	4	3	2	1
COMMON						B	A

One RS-232 connector.



Pin	Signal Direction	Signal Name	Signal Function
1	←	CD	Carrier Detect
2	→	TxD	Transmit Data
3	←	RxD	Receive Data
4	→	DTR	Data Terminal Ready
5	—	GND	Ground
6	←	DSR	Data Set Ready
7	→	CTS	Clear To Send
8	←	RTS	Request To Send
9	←	RI	Ring Indicator

Magnetic sensors



Command reference

Background

- The FCB can be controlled via serial interface either RS-232 or RS-485. Data format:
 - Baud Rate – 115200
 - Data Bits – 8
 - Parity – None
 - Stop Bits – 1
 - Flow Control – None
- Device address is set with the switch on PCB: SW1 – MSB of the address, SW5 – LSB. SW6 is intended for connecting termination resistor on RS-485 bus. Normally SW6 must be in OFF position
- Every command starts with 2-digit address from 01 to 31, or example “01R”
- Every command ends with <CR> symbol. <NL> symbol is ignored.
- Commands are case sensitive
- Device responses only in case of receiving its own address
- Device response starts with its 2-digit address and <SP> character, for example “01 OK”
- Every response ends with <CR> symbol
- Unrecognized commands are responded by “xx Fail”
- Data refresh rate for fan speed and magnetic field is 1 second.
- 100 ms minimum pause is recommended before sending next command after receiving a response from the previous command.

Command description

- xxR – read device status. Response “xx rpm1 rpm2 rpm3 rpm4 rpm5 field1 field2 err1 err2 L M” where:
 - xx – device address
 - rpm1 .. rpm5 fan rotation speeds, roots per minutes
 - field1 – magnetic field, measured by 1-st sensor, mT
 - field2 – magnetic field, measured by 2-nd sensor, mT
 - err1 – 1-st magnetic sensor error code, normally must be 0. If $err1 \neq 0$, field1 is invalid.
 - err2 – 2-nd magnetic sensor error code, normally must be 0. If $err2 \neq 0$, field2 is invalid.
 - L – L relay status: 0 – OFF, 1 – ON
 - M – M relay status: 0 – OFF, 1 – ON
- xxL0 – turn relay L OFF. Response “xx OK”
- xxL1 - turn relay L ON. Response “xx OK”
- xxM0 – turn relay M OFF. Response “xx OK”
- xxM1 - turn relay M ON. Response “xx OK”
- xxversion – reads HW and FW versions
- xxdiagnostic_mode_on – enters diagnostic mode
- xxboard_reset – resets board MCU
- xxremote_upgrade mode

Examples

- Command “01R”, response “01 5600 5456 5555 5432 5612 0.1 0.3 0 0 0 0”
- Command “01L1”, response “01 OK”
- Command “01version”, response “01 1.1 1.0”
- Command “01r”, response “01 Fail”
- Command “33R”, no response
- Command “05version”, response “05 1.1 1.2”

- Command “07diagnostic_mode_on”, response

07 OK

07 5400 5400 5399 5399 5401 104.6 113.2 0 0 0 0

07 5402 5401 5400 5401 5402 104.6 113.3 0 0 0 0

07 5399 5399 5399 5399 5399 104.6 113.2 0 0 0 0

Diagnostic mode reference

To enter diagnostic mode, send command `xxdiagnostic_mode_on`, where `xx` is the board address. Example: `08diagnostic_mode_on`.

In the diagnostic mode all commands are initiated by simply pressing a key.

- ‘ ‘ – SPACE toggle fans ON/OFF
- ‘0’ to ‘9’ – set rotation speed from 10% to 90% from full range
- ‘f’ – set full rotation speed
- ‘l’ – turn OFF relay L
- ‘L’ – turn ON relay L
- ‘m’ – turn OFF relay M
- ‘M’ – turn ON relay M
- ‘x’ – exit diagnostic mode
- ‘z’ – reset microcontroller and exit diagnostic mode

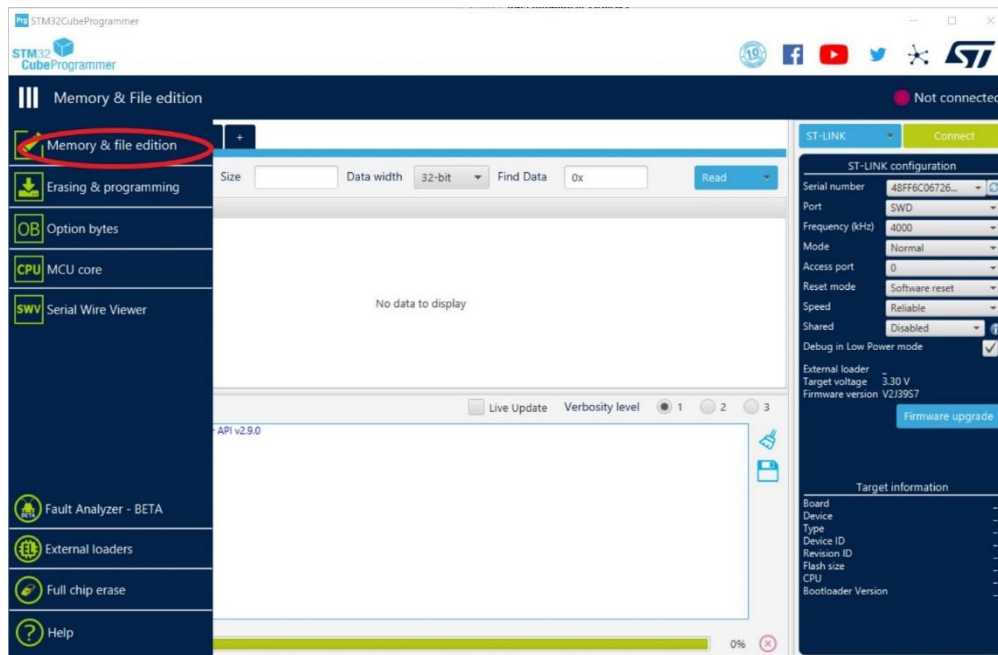
Firmware downloading

The firmware can be downloaded into the board using a hardware programmer. This section describes the procedure, which uses **ST-link V2 hardware programmer** and **STM32CubeProgrammer software**. This software is available for Windows, Mac and Linux platforms and can be downloaded from <https://www.st.com/en/development-tools/stm32cubeprog.html>. Downloading requires a registration.

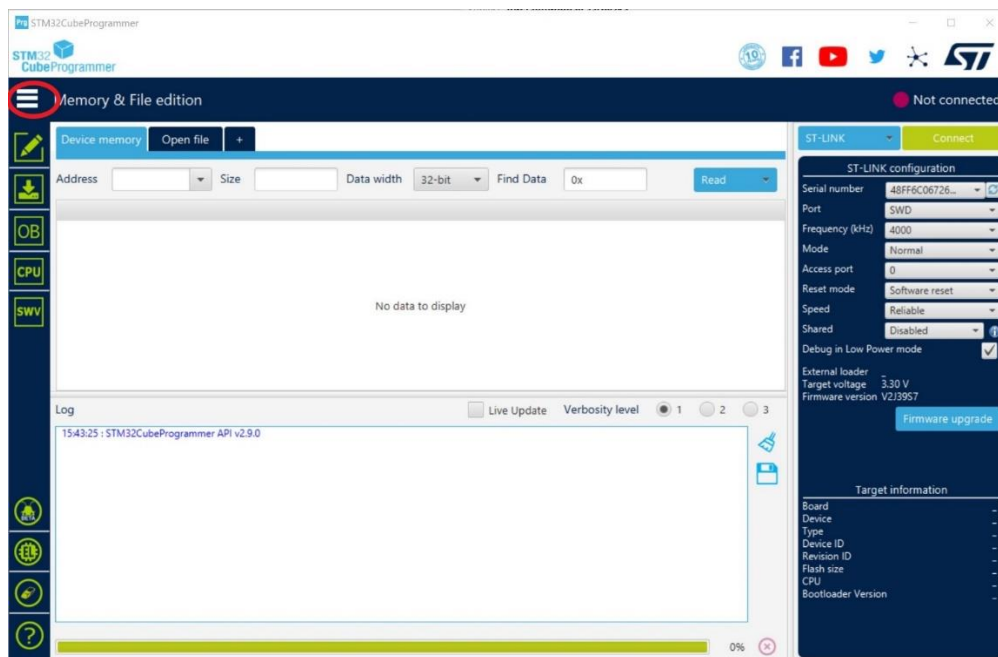


1. Install **STM32CubeProgrammer software** and necessary drivers in the laptop. **Warning:** *when programming FCB via laptop, disconnect the laptop from power supply and use only battery-powered laptop.*
2. **Warning:** *before connecting the hardware be sure that main and auxiliary power supplies for FCB board are switched off or disconnected.* Connect the cable from **ST-link V2 hardware programmer** to the board, then connect USB cable from programmer to the laptop.



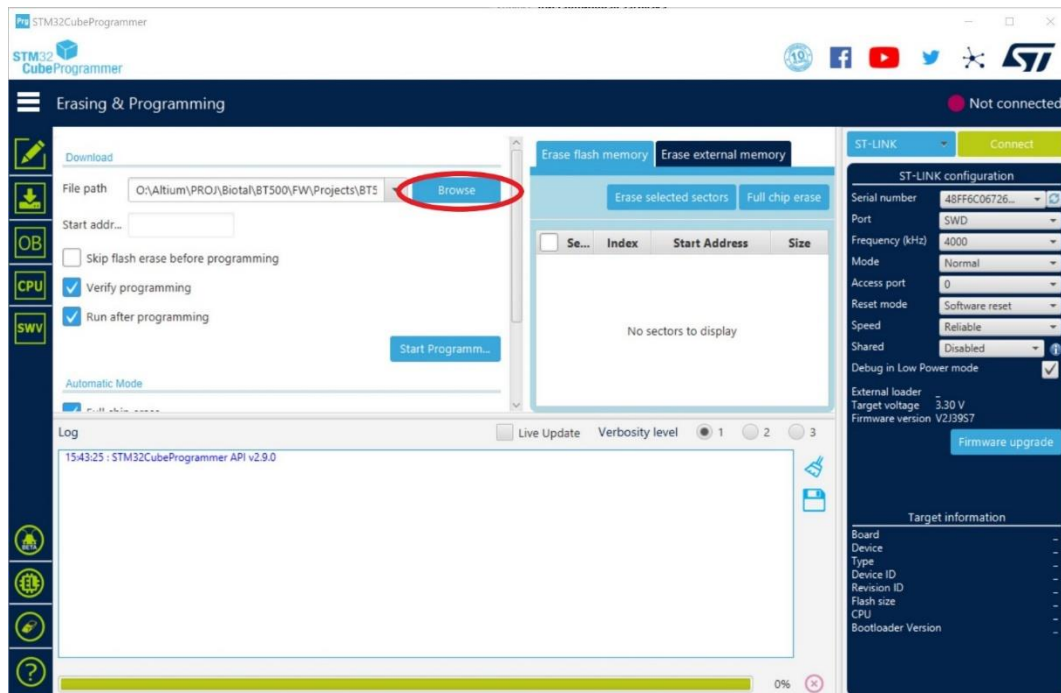


3. Power the board using auxiliary power supply
4. Start STM32CubeProgrammer software and click **Memory and File edition** icon

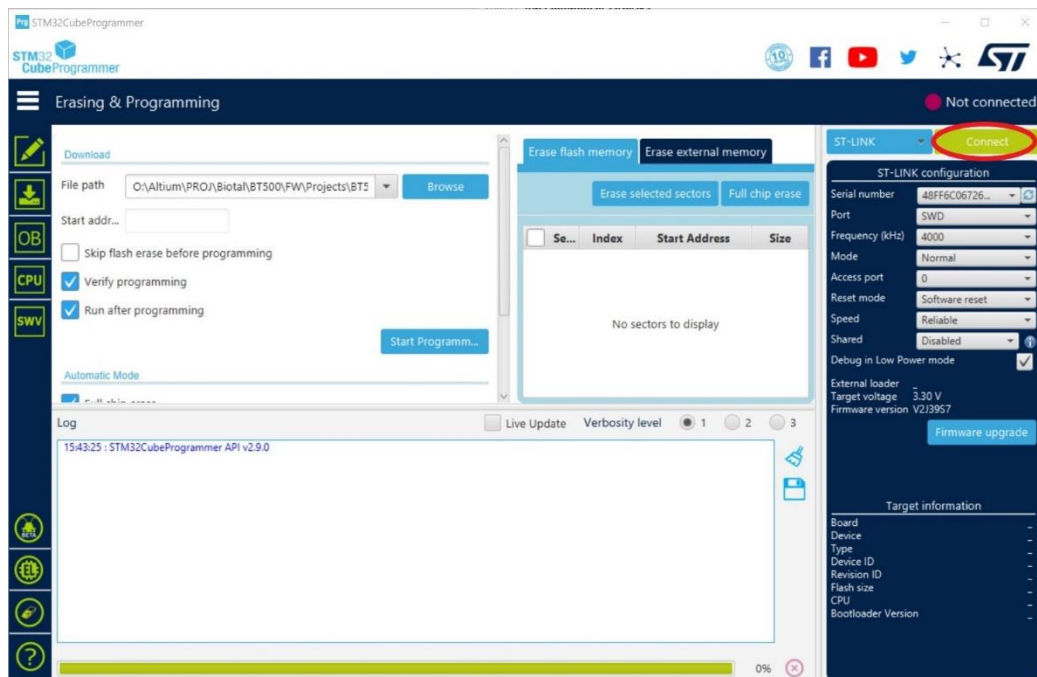


5. Click **Erasing & Programming** tab
6. Select **fc1_1_04_full.hex** file for downloading both bootloader and target firmware or select **fc1_openblt.srec** file to download only bootloader using **Browse** tab. If you download only bootloader, you will need the next step to download target firmware using serial port, see

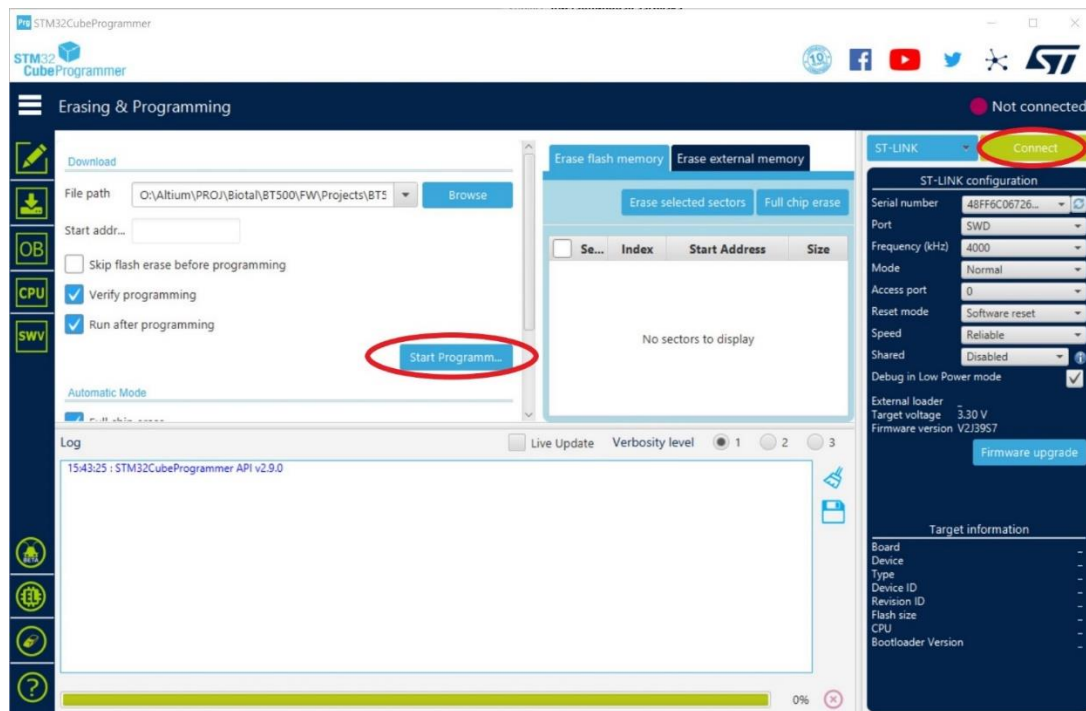
Remote upgrade mode section



7. The boxes **Verify programming** and **Run after programming** should be checked
8. Click on the green Connect button on the right corner



9. If connection is successful click the **Start programming** button



10. After successful programming un-power the board and disconnect programming cable.
11. When you then turn the board's power on, the channel 6 LED starts flashing rapidly for a moment, then slowly.

Remote upgrade mode

This mode allows to load a new firmware to the board remotely via serial interface RS-232 or RS-485. Implementation is based on the open-source project “OpenBLT Bootloader”

<https://www.feaser.com/openblt/doku.php?id=homepage>

Steps to prepare the board for remote upgrade for the first time:

12. Load the bootloader firmware to the board with the hardware programmer. After downloading the channel 6 LED starts flashing rapidly.
13. Start MicroBoot Utility on the host computer, select communication port and communication speed and select firmware file. When uploading the new firmware, the board's led 6 is flashing rapidly. After successful download the led begins flashing slowly.

In normal conditions to enter remote upgrade mode send command `xxremote_upgrade_mode`, where `xx` is the board address. Example: `08remote_upgrade_mode`.

In the remote upgrade mode, the board is waiting for connection to the host (channel 6 LED is flashing rapidly). Use Microboot utility to download the new firmware.